WSABI Summer Research Lab **Summer 2026**

Lecture 12: Empirical Bayes

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12.1 Which Batting Averages Will Persist?

At the end of the 2020 Covid-shortened MLB season, some players finished with unusually high or low batting averages in relatively few at-bats. Suppose our job is to use those 2020 results to predict 2021.

Player	2020 hits/AB	2020 AVG	Prediction for 2021?
A	9/20	.450	?
B	45/150	.300	?
C	70/250	.280	?
D	12/80	.150	?
E	90/300	.300	?

Which .300 hitter is more believable? Which hot start should we distrust? Which cold start should move back toward the pack?

For player i, let

$$X_i = \frac{H_i}{N_i}$$

be the observed batting average from H_i hits in N_i at-bats, and let μ_i be the player's underlying batting ability.

The raw estimate is

$$\widehat{\mu}_i^{\text{MLE}} = X_i.$$

But the leaderboard mixes real differences in player quality with sampling noise. A .390 average in 40 at-bats should move more than a .310 average in 300 at-bats.

By the end, we will build an estimator that beats the raw leaderboard at predicting next year.

12.2 Fake Data Learned From the League

Lecture 11 used fake data chosen from outside information. We chose a prior center and prior strength, then updated with observed wins and losses.

Empirical Bayes uses the same fake-data idea, but learns the fake data from the population. For batting averages, the league teaches us the prior fake hits and fake outs.

There are three broad strategies for estimating many related quantities:

- **No pooling**: estimate every player separately using X_i.
- **Complete pooling**: predict every player with one league mean.

- **Partial pooling**: move every player toward the league mean, but not all the way.

Partial pooling recognizes that players differ, while also recognizing that every player in this dataset belongs to a related league population.

Definition 12.1 (Empirical Bayes). *Empirical Bayes estimates the hyperparameters of a prior distribution from the observed population, then plugs those estimates into a Bayesian posterior calculation.*

Empirical Bayes is Bayesian updating with an estimated prior. The prior is not an external scouting report here. The prior is the estimated league distribution.

12.3 Beta–Binomial Empirical Bayes

A natural model for hit counts is

$$H_i \mid p_i \sim \text{Binomial}(N_i, p_i),$$
$$p_i \sim \text{Beta}(\alpha, \beta).$$

If (α, β) were known, the posterior mean would be

$$\mathbb{E}[p_i \mid H_i, N_i] = \frac{H_i + \alpha}{N_i + \alpha + \beta}.$$

Empirical Bayes estimates (α, β) from the league:

- $\widehat{\alpha}$ is league-learned fake hits.
- $\widehat{\beta}$ is league-learned fake outs.
- $\widehat{\alpha} + \widehat{\beta}$ is league-learned prior strength.
- $\widehat{\alpha}/(\widehat{\alpha} + \widehat{\beta})$ is the league-learned prior center.

The center comes from the league batting average; the strength comes from how much players appear to differ after accounting for sampling noise. One likelihood-based way to estimate these values is to choose (α, β) that make the observed hit counts most plausible after averaging over possible player abilities. The empirical-Bayes estimate is therefore

$$\widehat{p}_i^{\text{EB}} = \frac{H_i + \widehat{\alpha}}{N_i + \widehat{\alpha} + \widehat{\beta}}.$$

Actual hits plus learned fake hits, divided by actual at-bats plus learned fake at-bats.

Example 12.2 (A Hot Start in 20 At-Bats). Suppose a player has $H_i = 9$ hits in $N_i = 20$ at-bats, and the estimated league distribution is $\text{Beta}(60, 180)$. The raw batting average is $9/20 = 0.450$, while

$$\widehat{p}_i^{\text{EB}} = \frac{9 + 60}{20 + 60 + 180} = 0.265.$$

The short hot streak moves the estimate above the prior center of 0.250, but 20 at-bats provide little evidence for a true .450 hitter.

12.4 Did Shrinkage Predict Better?

The estimated-prior version of fake data has to earn its keep. Here, batting averages from the shortened 2020 season are used to predict 2021 batting averages. The 2021 data are not used to fit the empirical-Bayes model.

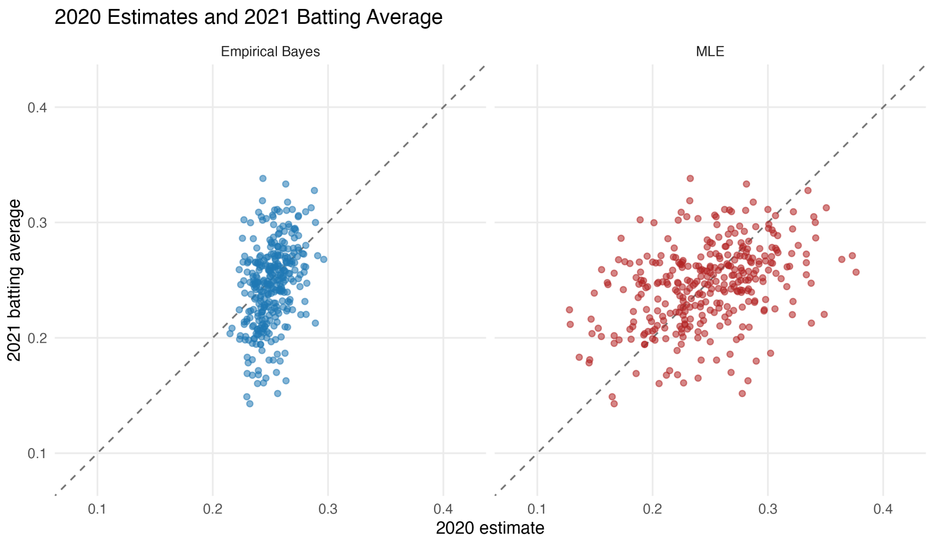

Figure 12.1: Raw and empirical-Bayes 2020 estimates against 2021 batting average.

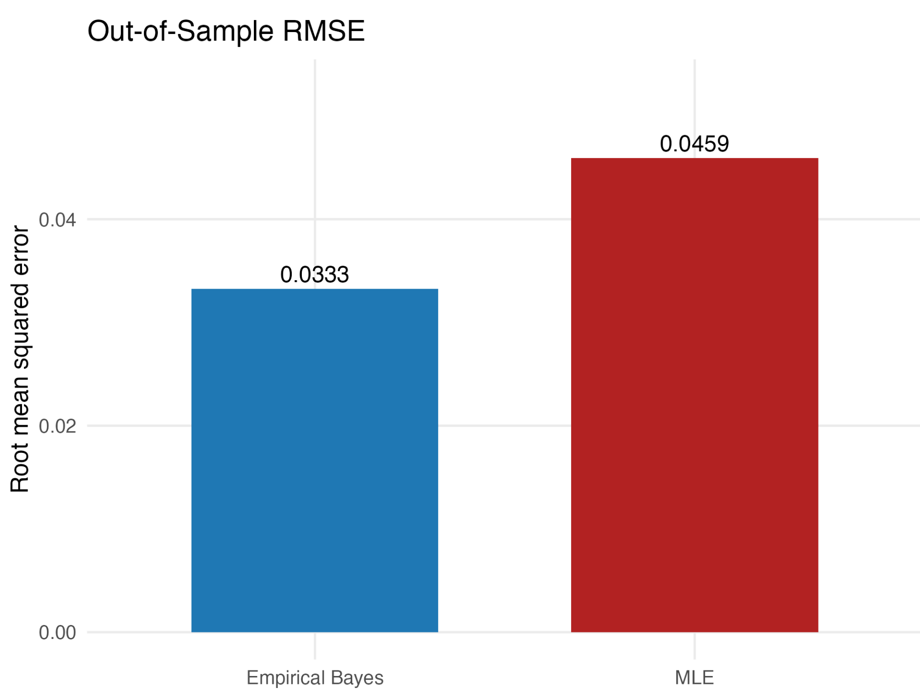

Figure 12.2: Out-of-sample RMSE against 2021 batting average.

12.5 How Much Does Each Player Move?

Let

$$\widehat{m} = \frac{\widehat{\alpha}}{\widehat{\alpha} + \widehat{\beta}}, \qquad \widehat{s} = \widehat{\alpha} + \widehat{\beta}.$$

Then the Beta–Binomial empirical-Bayes estimate can be written as

$$\widehat{p}_i^{\text{EB}} = \frac{N_i}{N_i + \widehat{s}} X_i + \frac{\widehat{s}}{N_i + \widehat{s}} \widehat{m}.$$

Large N_i gives more weight to the player's own average. Small N_i gives more weight to the league-learned fake data.

The largest moves come from players whose 2020 averages looked extreme relative to their number of at-bats.

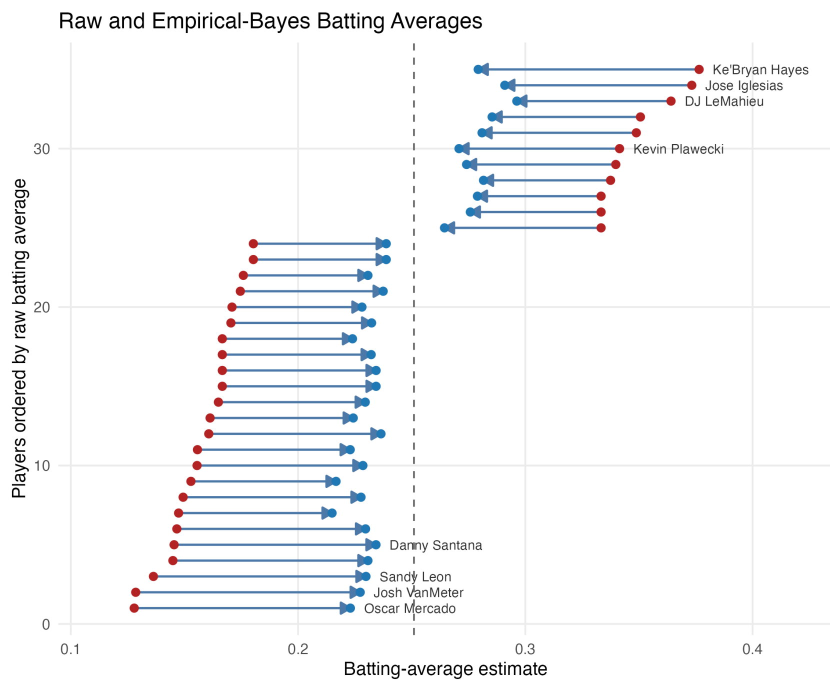

Figure 12.3: Raw and empirical-Bayes estimates for the 35 largest shrinkage adjustments, with selected players labeled.

The Normal–Normal model gives the same shrinkage geometry for estimates that are approximately Normal:

$$\begin{aligned} X_i \mid \mu_i &\sim \mathcal{N}(\mu_i, \sigma_i^2), \\ \mu_i &\sim \mathcal{N}(\mu, \tau^2). \end{aligned}$$

For a batting average, $\sigma_i^2 \approx C/N_i$, so players with fewer at-bats have noisier observed rates. The posterior mean is

$$\mathbb{E}[\mu_i \mid X_i] = \mu + \lambda_i(X_i - \mu), \quad \lambda_i = \frac{\tau^2}{\tau^2 + \sigma_i^2}.$$

After estimating the league distribution, the empirical-Bayes version is

$$\widehat{\mu}_i^{\text{EB}} = \widehat{\mu} + \frac{\widehat{\tau}^2}{\widehat{\tau}^2 + \sigma_i^2}(X_i - \widehat{\mu}).$$

This form is useful beyond batting averages because many estimates can be treated as approximately Normal once their standard errors are known.

12.6 Who Should Share Information?

Empirical Bayes borrows strength by pooling information across related players. The full group estimates the prior center $\widehat{\mu}$ and spread $\widehat{\tau}^2$; those estimates determine where each noisy batting average is pulled and how strongly it moves. This only makes sense when players are comparable enough to share a population model. However, this exchangeability assumption can fail:

- pitchers and position players may belong to different populations;
- rookies and veterans may need different prior centers;
- park, injury, role, and aging information may explain real heterogeneity.

The model is where we decide which players should share information. For example, instead of shrinking every player toward one common league mean, we could set up

$$\mu_i \sim \mathcal{N}(z_i^T\gamma, \tau^2),$$

where z_i contains characteristics such as position, experience, or park context. Players with similar z_i shrink toward similar targets, while τ^2 captures remaining differences not explained by those characteristics.

The same structure applies to free-throw rates, three-point shooting, quarterback efficiency, team scoring rates, and prospect projections. Partial pooling is extremely useful for both prediction and inference when related quantities are measured with substantial noise.

12.7 What Empirical Bayes Leaves Out

After estimating the hyperparameters, ordinary empirical Bayes treats them as *fixed*. Player-level uncertainty is then conditional on one fitted league distribution, $(\widehat{\mu}, \widehat{\tau}^2)$. A fully Bayesian model would also propagate uncertainty about (μ, τ^2), so empirical-Bayes intervals can be too narrow, especially with a small or noisy population.

That limitation is exactly what fully Bayesian hierarchical models address: estimate individual abilities, estimate the population distribution, and propagate uncertainty about both.

12.8 Takeaways

- A leaderboard contains both signal and sampling noise.
- Lecture 11 chose fake data externally; empirical Bayes learns fake data from related units.

- The Beta–Binomial estimate is actual hits plus league-learned fake hits.
- Shrinkage is stronger for noisier observations and weaker for larger samples.
- Out-of-sample prediction and exchangeability checks determine whether the pooling strategy is useful.
- Empirical Bayes is fast and useful, but it conditions on an estimated prior.

Appendix

A Beta–Binomial Marginal Likelihood

For the hit-count model, the player ability p_i is unobserved. The marginal likelihood averages over possible values of p_i:

$$\begin{aligned} p(H_i \mid N_i, \alpha, \beta) &= \int_0^1 p(H_i \mid N_i, p_i) p(p_i \mid \alpha, \beta)\, dp_i \\ &= \binom{N_i}{H_i} \frac{B(H_i + \alpha, N_i - H_i + \beta)}{B(\alpha, \beta)}. \end{aligned}$$

The empirical-Bayes estimates solve

$$(\widehat{\alpha}, \widehat{\beta}) = \arg\max_{\alpha,\beta>0} \sum_i \log p(H_i \mid N_i, \alpha, \beta).$$

B Normal–Normal Marginal Likelihood

To estimate the league distribution by likelihood, integrate out the unobserved ability μ_i:

$$\begin{aligned} p(X_i \mid \mu, \tau^2) &= \int p(X_i \mid \mu_i) p(\mu_i \mid \mu, \tau^2)\, d\mu_i \\ &= \mathcal{N}(X_i; \mu, \tau^2 + \sigma_i^2). \end{aligned}$$

The variance separates two sources of leaderboard spread:

$$\text{observed variation} = \text{real between-player variation} + \text{sampling noise}.$$

Assuming players are conditionally independent, the marginal likelihood multiplies their marginal densities. Taking logs turns that product into a sum, so the empirical-Bayes hyperparameter estimates solve

$$(\widehat{\mu}, \widehat{\tau}^2) = \arg\max_{\mu,\tau^2\geq 0} \sum_i \log \mathcal{N}(X_i; \mu, \tau^2 + \sigma_i^2).$$

C Normal–Normal Posterior Derivation

Combining the Normal likelihood and Normal prior gives

$$p(\mu_i \mid X_i) \propto \exp\left\{ -\frac{(X_i - \mu_i)^2}{2\sigma_i^2} - \frac{(\mu_i - \mu)^2}{2\tau^2} \right\}.$$

Collecting the terms involving μ_i produces a Normal posterior with precision $1/\sigma_i^2 + 1/\tau^2$ and mean

$$\frac{X_i/\sigma_i^2 + \mu/\tau^2}{1/\sigma_i^2 + 1/\tau^2} = \mu + \frac{\tau^2}{\tau^2 + \sigma_i^2}(X_i - \mu).$$

D Suggested Reading

- Bradley Efron and Carl Morris, "Stein's Paradox in Statistics," for the relationship between empirical Bayes and shrinkage.
- Andrew Gelman and Jennifer Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models*, on partial pooling.